

Example: Normal approximation to Binomial ^①

Given

X = number of bookings

$$X \sim \text{Binomial}(n=350, \pi=0.8)$$

The test

$$n\pi = (350)(0.8) = 280 \quad (> 5)$$

$$n(1-\pi) = (350)(0.2) = 70 \quad (> 5)$$

Show $X_a \sim N(\mu, \sigma^2)$ where $\mu = n\pi = 280$ and

$$\sigma^2 = n\pi(1-\pi) = 56 \text{ and}$$

$$\sigma = 7.4833$$

Calculate:

- (a) $P(X > 300)$ using approximation.
- (b) $P(X = 300)$ using approximation.
- (c) $P(X \leq 290)$ using approximation.

Solutions:

(2)

$$(a) P(X > 300) = 1 - P(X \leq 300)$$

$$\approx 1 - P(X_a < 300.5)$$

$$\approx 1 - P\left(\frac{X_a - \mu}{\sigma} < \frac{300.5 - 280}{7.4833}\right)$$

$$\approx 1 - P(Z < 2.7394)$$

$$\approx 1 - 0.9969$$

$$\approx \underline{0.0031} \rightarrow$$

$$(b) P(X = 300) \approx P(299.5 < X_a < 300.5)$$

$$\approx P(2.6058 < Z < 2.7394)$$

$$\approx P(Z < 2.7394) - P(Z < 2.6058)$$

$$\approx 0.9969 - 0.9954$$

$$\approx \underline{0.001483} \rightarrow$$

$$(c) P(X \leq 290) \approx P(X_a \leq 290.5)$$

$$\approx P(Z \leq 1.4031)$$

$$\approx \underline{0.9197} \rightarrow$$